

Avorix 0.5mg

Film Coated Tablets

1. NAME OF THE MEDICINAL PRODUCT

Avorix 0.5mg Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 0.532mg of Entecavir Monohydrate equivalent to 0.5mg Entecavir. For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film Coated Tablets.

White, round film-coated tablet debossed with “C21” on one side and plain on the other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Indicated for the treatment of chronic hepatitis B virus infection in adults with evidence of active viral replication and either evidence of persistent elevations in serum aminotransferases (ALT or AST) or histologically active disease.

The following points should be considered when initiating therapy with Entecavir:

- This indication is based on histologic, virologic, biochemical, and serologic responses in nucleoside-treatment-naïve and lamivudine-resistant adult patients with HBeAg- positive or HBeAg-negative chronic HBV infection with compensated liver disease.
- Virologic, biochemical, serologic, and safety data are available from a controlled study in adult subjects with chronic HBV infection and decompensated liver disease.
- Virologic, biochemical, serologic, and safety data are available for a limited number of adult subjects with HIV/HBV co-infection who have received prior lamivudine therapy.

4.2 Posology and method of administration

Compensated liver disease

Nucleoside-naïve patients: The recommended dose of Entecavir is 0.5 mg orally once daily with or without food.

Lamivudine-refractory patients: (i.e., history of hepatitis B viremia while receiving lamivudine therapy or known lamivudine resistance [LVDr, commonly called YMDD] mutations): The recommended dose is 1 mg once daily. Entecavir should be taken orally on an empty stomach (empty means at least 2 hours before and at least 2 hours after a meal).

Decompensated liver disease

The recommended dose for patients with decompensated liver disease is 1 mg once daily, which must be taken on an empty stomach (empty means at least 2 hours before and at least 2 hours after a meal).

Special Populations

Patients with renal impairment

Entecavir is predominantly eliminated by the kidney (see section 5.2). The clearance of entecavir decreases with impaired (decreasing) creatinine clearance (see section 5.2: Special Populations – Patients with renal impairment). Dosage adjustment of Entecavir is recommended for patients who have a creatinine clearance <50 mL/min, including those on hemodialysis or continuous ambulatory peritoneal dialysis (CAPD), as shown in Table 1.

Table 1: Recommended Dosage of Entecavir in Patients with Renal Impairment^a:

Creatinine Clearance (mL/min)	Usual Dose (0.5 mg once daily)	Lamivudine-Refractory or Decompensated Liver Disease (1 mg once daily)
30 – <50	0.5 mg every 48 hours	0.5 mg once daily OR 1 mg every 48 hours
10 – <30	0.5 mg every 72 hours	1 mg every 72 hours
<10	0.5 mg every 5–7 days	1 mg every 5–7 days
Hemodialysis ^b or CAPD	0.5 mg every 5–7 days	1 mg every 5–7 days

^a Do not split tablets.

^b On hemodialysis days, administer Entecavir after hemodialysis.

CAPD=continuous ambulatory peritoneal dialysis.

Patients with hepatic impairment

No dosage adjustment of Entecavir is required in patients with hepatic impairment.

Pediatric and adolescent patients

The safety and efficacy of Entecavir in patients <16 years of age have not been established.

Geriatric patients

No dosage adjustment of Entecavir based on age is required.

4.3 Contraindications

Entecavir is contraindicated in patients with previously demonstrated hypersensitivity to entecavir or any component of the product.

4.4 Special warnings and precautions for use

Drug-class-specific Warnings and Precautions

Lactic acidosis/hepatomegaly with steatosis

Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of nucleoside analogues alone or in combination with antiretrovirals.

Exacerbations of hepatitis after discontinuation of treatment

Acute exacerbation of hepatitis has been reported in patients who have discontinued HBV therapy, including therapy with Entecavir (see section 4.8).

The majority of post-treatment exacerbations appear to be self-limited. However, severe exacerbations, including fatalities, may occur. The causal relationship of these events to discontinuation of therapy is unknown. Hepatic function should be monitored at repeated intervals after discontinuation of therapy. If appropriate, resumption of HBV therapy may be warranted.

Product-specific Warnings and Precautions

Co-infection with HIV

Entecavir has not been evaluated in patients who are co-infected with human immunodeficiency virus (HIV) and HBV and are not concurrently receiving effective HIV treatment. Limited clinical experience suggests there is a potential for the development of HIV resistance if Entecavir is used to treat chronic hepatitis B infection in patients with untreated HIV infection (see section 5.1). Therefore, therapy with Entecavir is not recommended for HIV/HBV co-infected patients who are not also receiving highly active antiretroviral therapy (HAART). Entecavir has not been studied as a treatment for HIV infection and is not recommended for this use.

Patients with renal impairment

Dosage adjustment of Entecavir is recommended for patients with renal impairment (see section 4.2).

Liver transplant recipients

Limited data are available on the safety and efficacy of Entecavir in liver transplant recipients. In a single-arm, open-label study, patients who had HBV DNA less than 172 IU/mL at the time of transplant were treated with Entecavir 1 mg once daily post-transplant. The frequency and nature of adverse events in this study were consistent with those expected in patients who have received a liver transplant and the known safety profile of Entecavir. None of the 61 evaluable patients had virologic recurrence. Renal function should be carefully monitored before and during Entecavir therapy in liver transplant recipients receiving an immunosuppressant that may affect renal function, such as cyclosporine or tacrolimus (see section 4.2 and 5.2).

Patients with decompensated liver disease

A higher rate of serious hepatic adverse events (regardless of causality) has been observed in

patients with decompensated liver disease, in particular in those with Child-Turcotte- Pugh (CTP) class C disease, compared with rates in patients with compensated liver function. Also, patients with decompensated liver disease may be at higher risk for lactic acidosis and for specific renal adverse events such as hepatorenal syndrome. Therefore, clinical and laboratory parameters should be closely monitored in this patient population (see section 4.8).

Patient information

Patients should be advised that therapy with Entecavir has not been proven to reduce the risk of transmission of HBV and, therefore, appropriate precautions should still be taken.

Pediatric Use

Safety and effectiveness of Entecavir in pediatric patients below the age of 16 years have not been established.

4.5 Interactions with other medicinal products and other forms of interaction

Medicinal Products

Since entecavir is predominantly eliminated by the kidney (see section 5.2), coadministration of Entecavir with medicinal products that reduce renal function or compete for active tubular secretion may increase serum concentrations of either medicinal product. Coadministration of Entecavir with either lamivudine, adefovir dipivoxil or tenofovir disoproxil fumarate resulted in no significant drug interactions. The effects of coadministration of Entecavir with other medicinal products that are excreted renally or affect renal function have not been evaluated. Patients should be monitored closely for adverse events when Entecavir is coadministered with such medicinal products.

Food

Administration of entecavir with food decreased absorption by 18-20% (see section 4.2 and 5.2).

4.6 Fertility, pregnancy and lactation

There are no adequate and well-controlled studies in pregnant women. Entecavir should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

There are no data on the effect of entecavir on transmission of HBV from mother to infant.

Therefore, appropriate interventions should be used to prevent neonatal acquisition of HBV.

Entecavir is excreted in the milk of rats. It is not known whether it is excreted in human milk. Mothers should be instructed not to breast-feed if they are taking Entecavir.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Dizziness, fatigue and somnolence are common side effects which may impair the ability to drive and use machines.

4.8 Undesirable effects

Adverse reactions considered at least possibly related to treatment with entecavir are listed by body system organ class. Frequency is defined as very common; common; uncommon.

Nucleoside naïve patients

<i>Psychiatric disorders:</i>	uncommon: insomnia
<i>Nervous system disorders:</i>	common: headache uncommon: dizziness, somnolence
<i>Gastrointestinal disorders:</i>	uncommon: nausea, diarrhea, dyspepsia, vomiting
<i>General disorders and administration site conditions:</i>	common: fatigue

Lamivudine-refractory patients

<i>Nervous system disorders:</i>	common: headache
<i>Gastrointestinal disorders:</i>	common: diarrhea, dyspepsia
<i>General disorders and administration site conditions:</i>	common: fatigue

Laboratory Findings

Table 2: Selected Laboratory Abnormalities Reported During Treatment

Test	Nucleoside naïve patients	Lamivudine-refractory patients
ALT >10 × ULN and >2 × baseline	2%	2%
ALT >3 × baseline	5%	4%
ALT >2 × baseline and total bilirubin >2 × ULN and >2 × baseline	<1%	<1%
Albumin <2.5 g/dL	<1%	0
Amylase >3 × baseline	2%	2%
Lipase >3 × baseline	12%	18%
Platelets <50,000/mm ³	<1%	<1%

ULN=upper limit of normal.

Among Entecavir-treated patients, on-treatment ALT elevations $>10 \times \text{ULN}$ and $>2 \times$ baseline generally resolved with continued treatment. A majority of these exacerbations were associated with a $\geq 2 \log_{10}/\text{mL}$ reduction in viral load that preceded or coincided with the ALT elevation. Periodic monitoring of hepatic function is recommended during treatment.

Exacerbations of Hepatitis after Discontinuation of Treatment

Acute exacerbations of hepatitis have been reported in patients who have discontinued anti-HBV therapy, including therapy with Entecavir. The frequency of exacerbation of hepatitis or ALT flare (defined as ALT $>10 \times \text{ULN}$ and $2 \times$ the patient's reference level) during off-treatment follow-up is presented in Table 3.

Table 3: Exacerbations of Hepatitis During Off-Treatment Follow-up

	Patients with ALT Elevations $>10 \times \text{ULN}$ and $>2 \times \text{Reference}^a$	
	Entecavir	Lamivudine
Nucleoside naïve	28/476 (6%)	43/417 (10%)
HBeAg-positive	4/174 (2%)	13/147 (9%)
HBeAg-negative	24/302 (8%)	30/270 (11%)
Lamivudine-refractory	6/52 (12%)	0/16

^a Reference is the minimum of the baseline or last measurement at end of dosing. Median time to off-treatment exacerbation was 23 weeks for Entecavir-treated patients and 10 weeks for lamivudine-treated patients.

Decompensated liver disease

Additional adverse reactions observed in Entecavir treated patients in a study in which Entecavir 1 mg/day was compared with adefovir dipivoxil in patients with chronic hepatitis B infection and decompensated liver disease include decrease in blood bicarbonate (2%) and renal failure ($<1\%$). The on-study cumulative death rate was 23% (23/102), and causes of death were generally liver-related, as expected in this population. The on-study cumulative rate of hepatocellular carcinoma (HCC) was 12% (12/102).

Laboratory test abnormalities: none had ALT elevations both >10 times ULN and >2 times baseline, and 1% of patients had ALT elevations >2 times baseline together with total bilirubin >2 times ULN and >2 times baseline. Albumin levels $<2.5 \text{ g/dL}$ occurred in 30% of patients, lipase levels >3 times baseline in 10%, and platelets $<50,000/\text{mm}^3$ in 20%.

Patients Co-infected with HIV

Patients co-infected with HBV and HIV who experienced recurrence of HBV viremia while receiving a lamivudine-containing highly active antiretroviral regimen were treated with their lamivudine-containing regimen (lamivudine dose, 300 mg/day) and either Entecavir 1 mg once daily or placebo. After therapy, the adverse event and laboratory abnormality profiles were similar for the Entecavir and placebo treatment groups. Entecavir has not been evaluated in HIV/HBV co-infected patients who are not concurrently receiving effective IV treatment.

Postmarketing Experience

The principal outcome events assessed during a study were overall malignant neoplasms, liver-related HBV disease progression, HCC, non-HCC malignant neoplasms, and death. The study showed that Entecavir was not significantly associated with an increased risk of malignant neoplasms compared to other standard-of-care HBV nucleos(t)ides, as assessed by either the composite endpoint of overall malignant neoplasms or the individual endpoint of non-HCC malignant neoplasms. The most commonly reported malignancy in both the Entecavir and non-ETV groups was HCC followed by gastrointestinal malignancies. The data also showed that long-term Entecavir use was not associated with a lower occurrence of HBV disease progression or a lower rate of death overall compared to other HBV nucleos(t)ides. The principal clinical outcome event assessments are shown in Table 4.

Table 4: Principal Analyses of Time to Adjudicated Events - Randomized Treated Subjects

Endpoint ^c	Entecavir	Non ETV	Hazard Ratio [Entecavir:Non-ETV] (CI ^a)
Primary Endpoints			
Overall malignant neoplasm	331	337	0.93 (0.800, 1.084)
Liver-related HBV disease progression	350	375	0.89 (0.769, 1.030)
Death	238	264	0.85 (0.713, 1.012)
Secondary Endpoints			
Non-HCC malignant neoplasm	95	81	1.10 (0.817, 1.478)
HCC	240 ^b	263	0.87 (0.727, 1.032)

Analyses were stratified by geographic region and prior HBV nucleos(t)ide experience.

^a 95.03% CI for overall malignant neoplasm, death, and liver-related HBV disease progression; 95% CI for non-HCC malignant neoplasm and HCC.

^b One subject had a pre-treatment HCC event and was excluded from the analysis.

^c Overall malignant neoplasm is a composite event of HCC or non-HCC malignant neoplasm. Liver-related HBV disease progression is a composite event of liver-related death, HCC, or non-HCC HBV disease progression.

CI = confidence interval

Adverse Reactions from Postmarketing Spontaneous Reports

Immune system disorders: Anaphylactoid reaction.

Metabolism and nutrition disorders: Lactic acidosis has been reported, often in association with hepatic decompensation, other serious medical conditions, or drug exposures. Patients with decompensated cirrhosis may be at higher risk for lactic acidosis.

Hepatobiliary disorders: Increased transaminases.

Skin and subcutaneous tissue disorders: Alopecia, rash.

Because reports are voluntary from a population of unknown size, an estimate of frequency cannot be made.

4.9 Overdose

There is no experience of Entecavir overdose reported in patients. Healthy subjects who received up to 20 mg/day for up to 14 days, and single doses up to 40 mg had no unexpected adverse events. If overdose occurs, the patient must be monitored for evidence of toxicity and given standard supportive treatment as necessary.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: antivirals for systemic use, nucleoside and nucleotide reverse transcriptase inhibitors

ATC code: J05AF10

Mechanism of action

Entecavir is a guanosine nucleoside analogue with potent and selective activity against HBV polymerase. It is phosphorylated to the active triphosphate (TP) form, which has an intracellular half-life of 15 hours. Intracellular TP levels are directly related to extracellular entecavir concentrations, with no significant accumulation beyond initial plateau levels. By competing with the natural substrate deoxyguanosine-TP, entecavir-TP inhibits all 3 functional activities of the viral polymerase: (1) priming of the HBV polymerase, (2) reverse transcription of the negative strand from the pregenomic messenger RNA, and (3) synthesis of the positive strand HBV DNA. The entecavir-TP K_i for HBV DNA polymerase is 1.2 nM. Entecavir-TP is a weak inhibitor of cellular DNA polymerases α , β , and δ with K_i values of 18 to 40 μ M. In addition, high exposures of entecavir-TP and entecavir had no relevant adverse effects on γ polymerase ($K_i > 160 \mu$ M) or mitochondrial DNA synthesis in HepG2 cells.

Antiviral activity

Entecavir inhibited HBV DNA synthesis (50% reduction, EC₅₀) at a concentration of 0.004 μ M in human HepG2 cells transfected with wild-type HBV. The median EC₅₀ value for entecavir against lamivudine-resistant HBV (rtM204V, rtL180M) was 0.026 μ M (range 0.010-0.059 μ M).

A comprehensive analysis of the inhibitory activity of entecavir against a panel of laboratory and clinical HIV-1 isolates using a variety of cells and assay conditions yielded EC₅₀s ranging from 0.026 to $>10 \mu$ M. Inhibitory activity at concentrations $\sim 1 \mu$ M was only observed when cell culture assay conditions were modified in a way that decreased the level of virus used to initiate infection. In cell culture, entecavir selected for an M184I substitution at micromolar concentrations, confirming inhibitory pressure at high entecavir concentrations. HIV variants containing the M184V substitution showed loss of susceptibility to entecavir.

Resistance in cell culture

Compared to activity against wild-type HBV, there is an 8-fold reduction in susceptibility to entecavir in cell-based studies when lamivudine and telbivudine resistance-associated

substitutions rtM204I/V ± rtL180M (LVDr) are present. At extracellular concentrations representative of plasma levels achieved with 1-mg dosing, intracellular entecavir-TP levels would be expected to surpass those needed to inhibit the enzyme activity of lamivudine-resistant HBV polymerases. Recombinant viruses encoding adefovir-resistant substitutions at either rtN236T or rtA181V remained fully susceptible to entecavir.

5.2 Pharmacokinetic properties

Absorption

Entecavir was rapidly absorbed with peak plasma concentrations occurring between 0.5 and 1.5 hours. There was a dose-proportionate increase in peak plasma concentration ($C_{\max}$) and area under the concentration-time curve (AUC) values following multiple doses ranging from 0.1 to 1 mg. Steady-state was achieved after 6-10 days of once-daily dosing with approximately 2-fold accumulation. $C_{\max}$ and trough plasma concentration (C_{trough}) at steady-state were 4.2 and 0.3 ng/mL, respectively, for a 0.5-mg dose, and 8.2 and 0.5 ng/mL, respectively, for a 1-mg dose.

Oral administration of entecavir 0.5 mg with a standard high-fat meal (945 kcal, 54.6 g fat) or a light meal (379 kcal, 8.2 g fat) resulted in a minimal delay in absorption (1.0-1.5 hours fed vs. 0.75 hour fasted), a decrease in $C_{\max}$ of 44-46%, and a decrease in AUC of 18-20%.

Distribution

The estimated volume of distribution for entecavir was in excess of total body water, suggesting that it has good penetration into tissues. Protein binding to human serum protein *in vitro* was approximately 13%.

Metabolism and elimination:

Entecavir is not a substrate, inhibitor, or inducer of the CYP450 enzyme system. At concentrations approximately 10,000-fold higher than those obtained in humans, entecavir inhibited none of the major human CYP450 enzymes 1A2, 2C9, 2C19, 2D6, 3A4, 2B6, and 2E1. At concentrations approximately 340-fold higher than those observed in humans, entecavir did not induce the human CYP450 enzymes 1A2, 2C9, 2C19, 3A4, 3A5, and 2B6. Following administration of ^{14}C -entecavir in humans and rats, no oxidative or acetylated metabolites and minor amounts of phase II metabolites (glucuronide and sulfate conjugates) were observed.

After reaching peak levels, entecavir plasma concentrations decreased in a bi-exponential manner with a terminal elimination half-life of approximately 128-149 hours. The observed drug accumulation index is approximately 2-fold with once-daily dosing, suggesting an effective accumulation half-life of about 24 hours.

Entecavir is predominantly eliminated by the kidney with urinary recovery of unchanged drug at steady-state ranging from 62% to 73% of the dose. Renal clearance is independent of dose and ranges between 360 and 471 mL/min suggesting that entecavir undergoes both glomerular filtration and net tubular secretion.

Special populations:

Patients with renal impairment:

Based on the results of a study of the pharmacokinetics of a single 1-mg dose of entecavir in patients with renal impairment (Table 5), dosage adjustment is recommended for patients who have a creatinine clearance <50 mL/min.

Table 5: Pharmacokinetic Parameters of Entecavir in Subjects with Selected Degrees of Renal Impairment

	Renal Function Group					
	Baseline Creatinine Clearance (mL/min)				Severe Managed with Hemodialysis (n=6)	Severe Managed with CAPD (n=4)
	Unimpaired >80 (n=6)	Mild >50 - ≤80 (n=6)	Moderate 30-50 (n=6)	Severe <30 (n=6)		
C _{max} (ng/mL) (CV%)	8.1 (30.7)	10.4 (37.2)	10.5 (22.7)	15.3 (33.8)	15.4 (56.4)	16.6 (29.7)
AUC _(0-T) (ng•hr/mL) (CV)	27.9 (25.6)	51.5 (22.8)	69.5 (22.7)	145.7 (31.5)	233.9 (28.4)	221.8 (11.6)
CLR (mL/min) (SD)	383.2 (101.8)	197.9 (78.1)	135.6 (31.6)	40.3 (10.1)	NA	NA
CLT/F (mL/min) (SD)	588.1 (153.7)	309.2 (62.6)	226.3 (60.1)	100.6 (29.1)	50.6 (16.5)	35.7 (19.6)

CLR=renal clearance; CLT/F=apparent oral clearance; CAPD=continuous ambulatory peritoneal dialysis.

A 4-hour period of hemodialysis removed approximately 13% of the dose, and 0.3% was removed by CAPD.

Patients with hepatic impairment:

Pharmacokinetic parameters in patients with hepatic impairment were similar to those in patients with normal hepatic function.

Geriatric

The pharmacokinetic profile of entecavir does not differ by age.

Gender/Race

The pharmacokinetic profile of entecavir does not differ by gender or race.

Liver transplant recipients: Entecavir exposure in HBV-infected liver transplant recipients on a stable dose of cyclosporine A (n=5) or tacrolimus (n = 4) was approximately 2-fold the exposure in healthy subjects with normal renal function. Altered renal function contributed to the increase in entecavir exposure in these patients (see section 4.4).

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Tablet core:

Lactose Monohydrate, Microcrystalline Cellulose, Crospovidone, Povidone, Magnesium Stearate

Film-coat:

Opadry Complete Film Coating System YS-1-18027-A White [*Components: Titanium Dioxide, Hypromellose (6mPas), Hypromellose(3mPas), Macrogol, Polysorbate 80*].

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

Please refer to the expiry date on the product labels.

6.4 Special precautions for storage

Store below 30°C. Store in the original package.

6.5 Nature and contents of container

Box of 1 x 10's, 3 x 10's, 6 x 10's, 10 x 10's, 50 x 10's and 100 x 10's tablets in Alu/ Alu blisters in outer cartons. Not all pack sizes may be marketed.

6.6 Special precautions for disposal

No special requirement.

7. Manufacturer

Manufactured by:

Novugen Pharma Sdn. Bhd.
No.27, Jalan Lengku Teknologi 2,
Taman Teknologi Enstek Fasa 1,
71760 Bandar Baru Enstek,
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